

2021-Ph-H1-Q20

Section: Electricity

Topic: Monitoring and Measuring a.c.

A sinusoidal voltage has a peak of 10 V. What is the RMS value?

Solution:

$$V_{\text{rms}} = \frac{V_{\text{peak}}}{\sqrt{2}} = \frac{10}{\sqrt{2}} \approx 7.1 \text{ V}$$

Answer: B

Guidance for Students:

RMS (root mean square) is the effective voltage. Multiply by $\sqrt{2}$ to go between RMS and peak.

Revision Tip:

Know:

$$V_{\text{rms}} = \frac{V_{\text{peak}}}{\sqrt{2}} \approx 0.707V_{\text{peak}}$$
